N-Channel Power MOSFET

60V, 11A, 90mΩ

FEATURES

- 100% UIS and Rg tested
- Logic-level gate drive
- Fast switching
- RoHS Compliant
- Halogen-Free according to IEC 61249-2-21

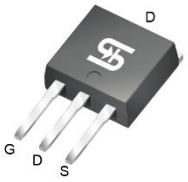
APPLICATIONS

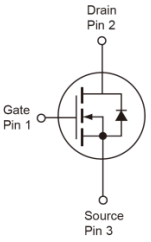
- DC-DC Converters
- Solenoid and Motor Drivers

PRODUCT SUMMARY			
PARAMETER		**VALUE**	**UNIT**
V_{DS}		60	V
$R_{DS(on)}$ (max)	V_{GS} = 10V	90	mΩ
	V_{GS} = 4.5V	100	
Q_g	V_{GS} = 10V	9.5	nC

Pb-Free RoHS COMPLIANT HALOGEN FREE

TO-251S (IPAK SL)





ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)				
PARAMETER		**SYMBOL**	**LIMIT**	**UNIT**
Drain-Source Voltage		V_{DS}	60	V
Gate-Source Voltage		V_{GS}	±20	V
Continuous Drain Current (Note 1)	T_C = 25°C	I_D	11	A
	T_C = 100°C		7	
Pulsed Drain Current (Note 2)		I_{DM}	44	A
Single Pulse Avalanche Current (Note 3)		I_{AS}	7	A
Single Pulse Avalanche Energy (Note 3)		E_{AS}	25	mJ
Total Power Dissipation	T_C = 25°C	P_D	25	W
Operating Junction and Storage Temperature Range		T_J, T_{STG}	- 55 to +150	°C

THERMAL RESISTANCE			
PARAMETER	**SYMBOL**	**MAXIMUM**	**UNIT**
Thermal Resistance – Junction to Case	$R_{\Theta JC}$	5	°C/W
Thermal Resistance – Junction to Ambient	$R_{\Theta JA}$	62	°C/W

Note: $R_{\Theta JA}$ is the sum of the junction-to-case and case-to-ambient thermal resistances. The case-thermal reference is defined at the solder mounting surface of the drain pins. $R_{\Theta JC}$ is guaranteed by design while $R_{\Theta CA}$ is determined by the user's board design

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ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)						
PARAMETER	**CONDITIONS**	**SYMBOL**	**MIN**	**TYP**	**MAX**	**UNIT**
Static						
Drain-Source Breakdown Voltage	V_{GS} = 0V, I_D = 250μA	BV_{DSS}	60	--	--	V
Gate Threshold Voltage	V_{GS} = V_{DS}, I_D = 250μA	$V_{GS(TH)}$	1.2	1.8	2.5	V
Gate-Source Leakage Current	V_{GS} = ±20V, V_{DS} = 0V	I_{GSS}	--	--	±100	nA
Drain-Source Leakage Current	V_{GS} = 0V, V_{DS} = 60V	I_{DSS}	--	--	1	μA
	V_{GS} = 0V, V_{DS} = 48V T_J = 125°C		--	--	10	
Drain-Source On-State Resistance (Note 4)	V_{GS} = 10V, I_D = 6A	$R_{DS(on)}$	--	76	90	mΩ
	V_{GS} = 4.5V, I_D = 3A		--	87	100	
Forward Transconductance (Note 4)	V_{DS} = 10V, I_D = 3A	g_{fs}	--	4	--	S
Dynamic						
Total Gate Charge	V_{GS} = 10V, V_{DS} = 48V, I_D = 6A	Q_g	--	9.5	--	nC
Gate-Source Charge		Q_{gs}	--	2	--	
Gate-Drain Charge		Q_{gd}	--	1.4	--	
Input Capacitance	V_{GS} = 0V, V_{DS} = 30V, f = 1.0MHz	C_{iss}	--	553.4	--	pF
Output Capacitance		C_{oss}	--	34.4	--	
Reverse Transfer Capacitance		C_{rss}	--	27	--	
Gate Resistance	f = 1.0MHz	R_g	--	2	--	Ω
Switching (Note 5)						
Turn-On Delay Time	V_{GS} = 10V, V_{DS} = 30V, I_D = 1A, R_G = 3.3Ω	$t_{d(on)}$	--	6.7	--	ns
Rise Time		t_r	--	2.8	--	
Turn-Off Delay Time		$t_{d(off)}$	--	17.1	--	
Fall Time		t_f	--	1.8	--	
Source-Drain Diode						
Diode Forward Voltage (Note 4)	V_{GS} = 0V, I_S = 1A	V_{SD}	--	--	1.2	V
Reverse Recovery Time	I_S = 2A, V_{GS} = 30V di/dt = 100A/μs	t_{rr}	--	12.5	--	ns
Reverse Recovery Charge		Q_{rr}	--	7.7	--	nC

Notes:

1. Limited by maximum junction temperature.
2. Repetitive Rating : Pulsed width limited by maximum junction temperature.
3. L = 1mH , V_{GS} = 10V, R_G = 25Ω , Starting T_J = 25°C.
4. Pulse test: Pulse Width ≤ 300μs, duty cycle ≤ 2%.
5. Switching time is essentially independent of operating temperature.

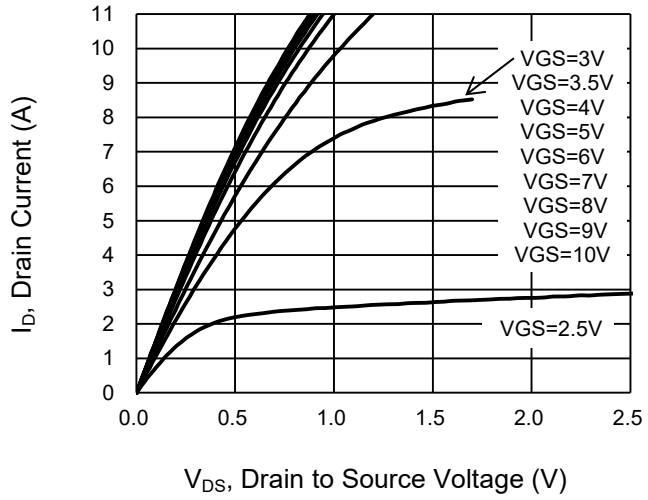
ORDERING INFORMATION

ORDERING CODE	PACKAGE	PACKING
TSM900N06CH X0G	TO-251S	75pcs / Tube

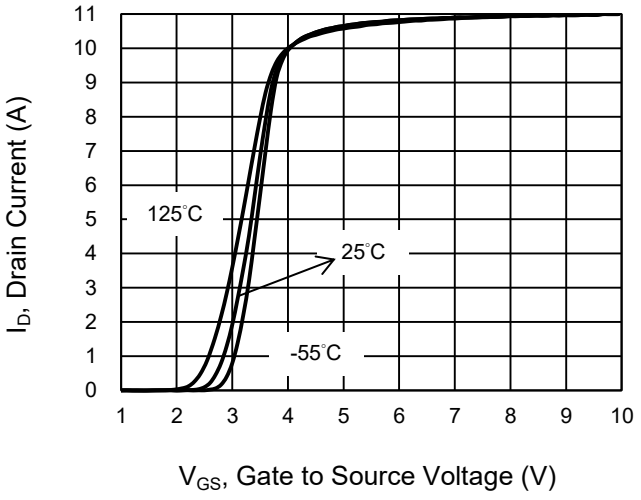
CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

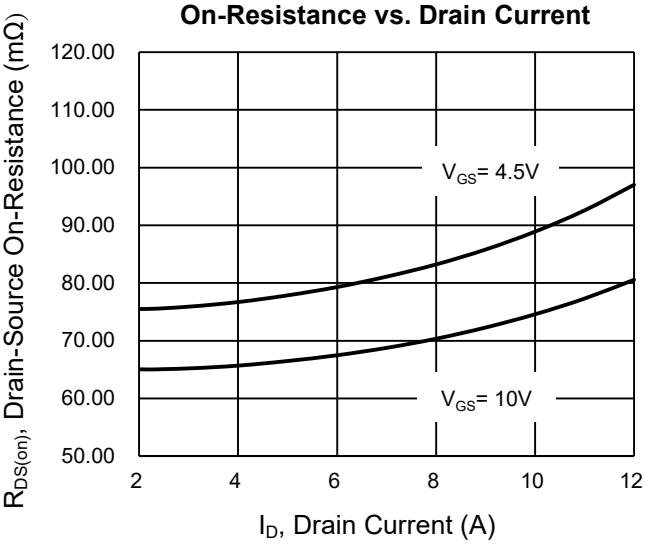
Output Characteristics



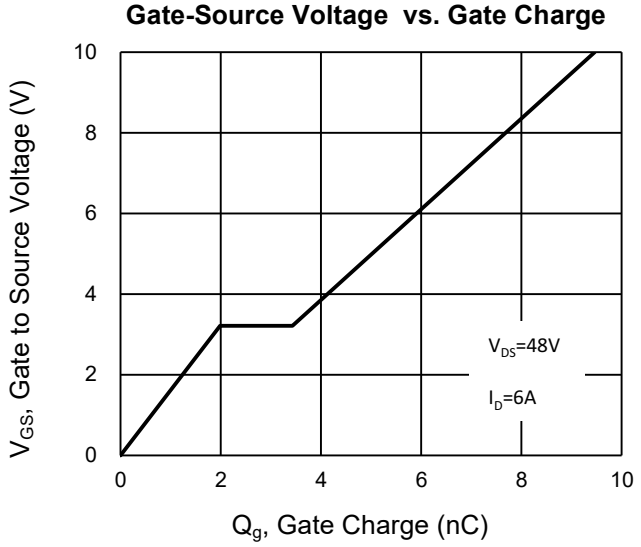
Transfer Characteristics



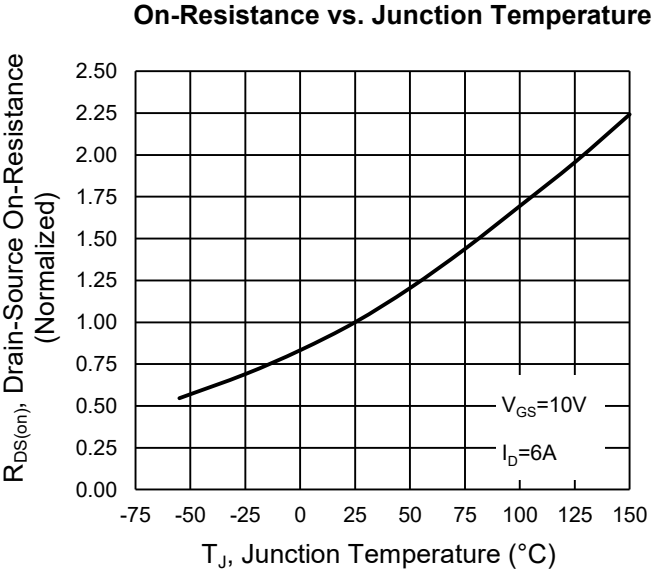
On-Resistance vs. Drain Current



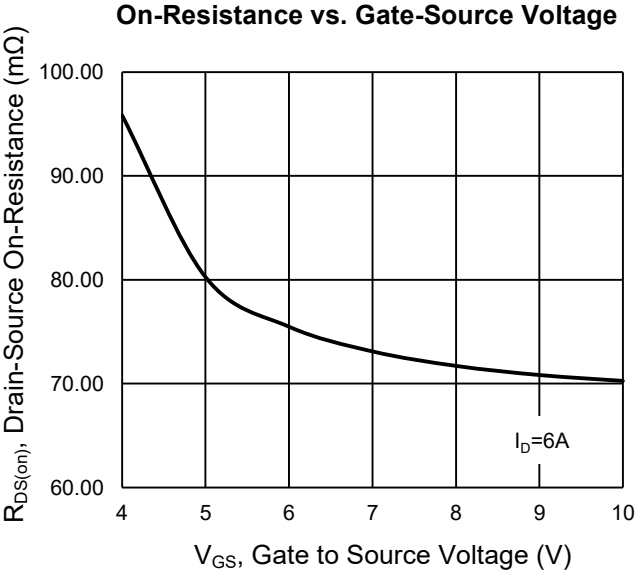
Gate-Source Voltage vs. Gate Charge



On-Resistance vs. Junction Temperature



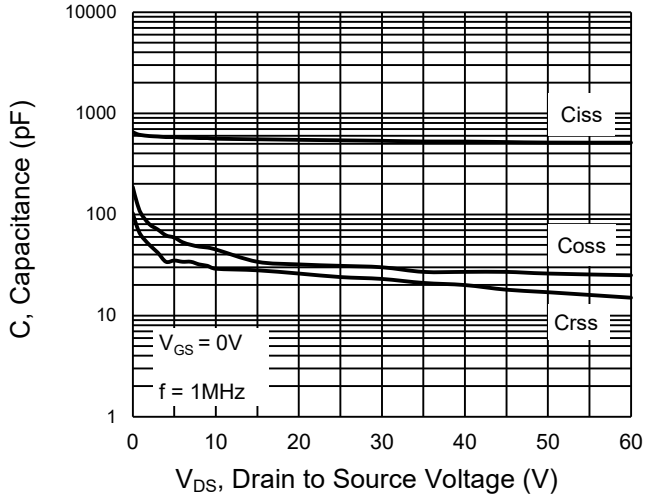
On-Resistance vs. Gate-Source Voltage



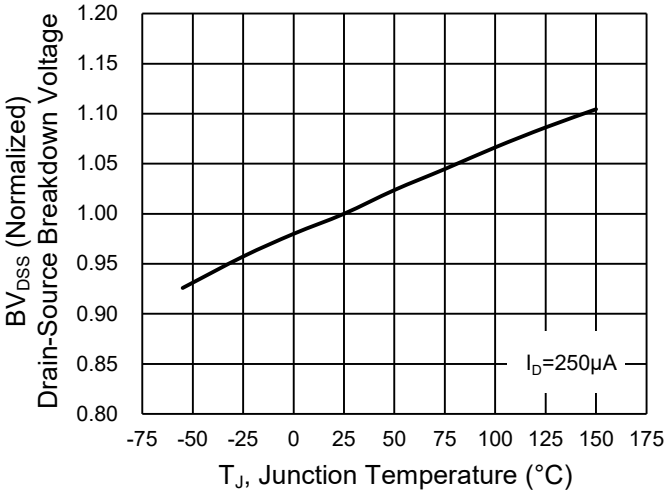
CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

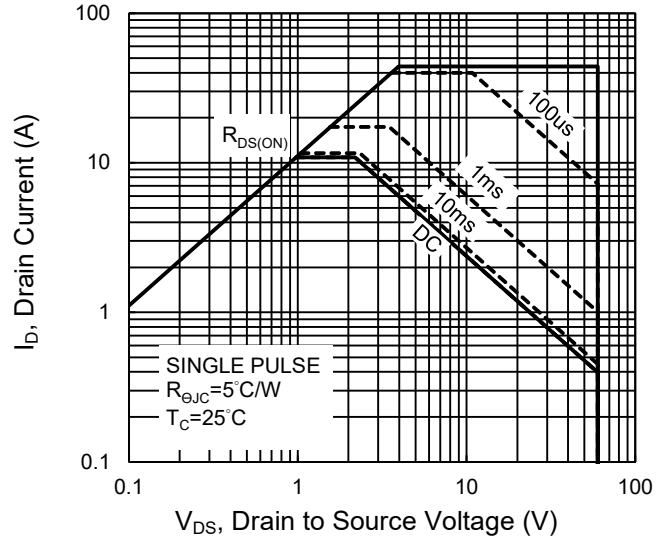
Capacitance vs. Drain-Source Voltage



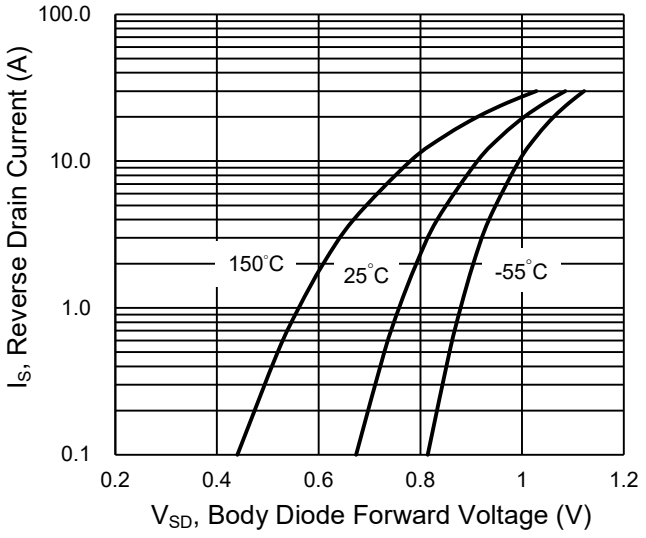
BV_{DSS} vs. Junction Temperature



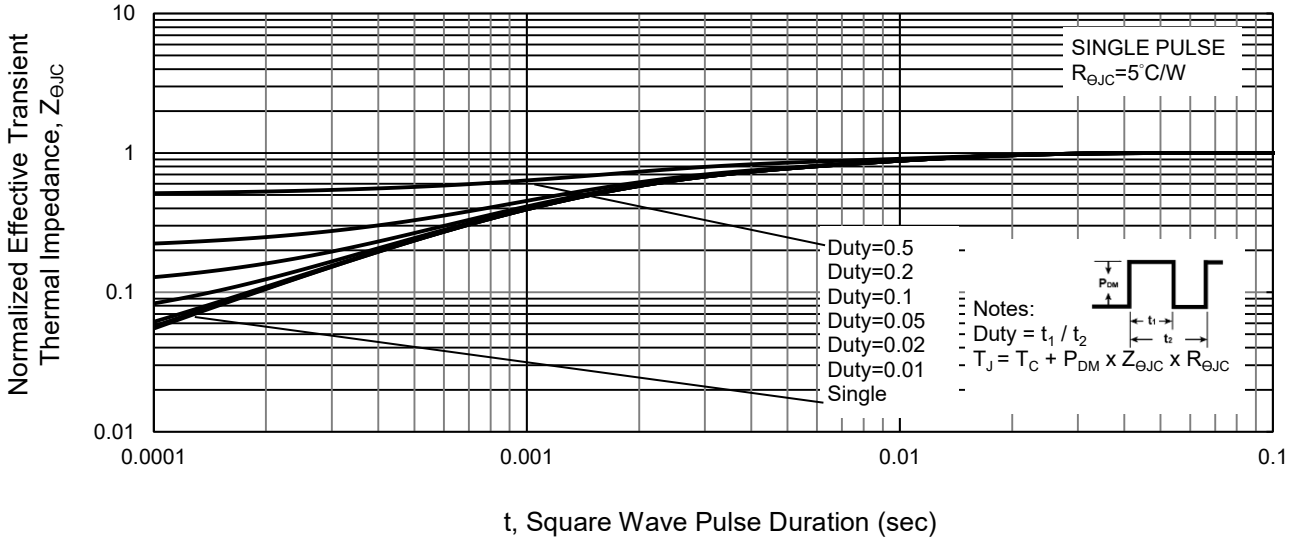
Maximum Safe Operating Area, Junction-to-Case



Source-Drain Diode Forward Current vs. Voltage



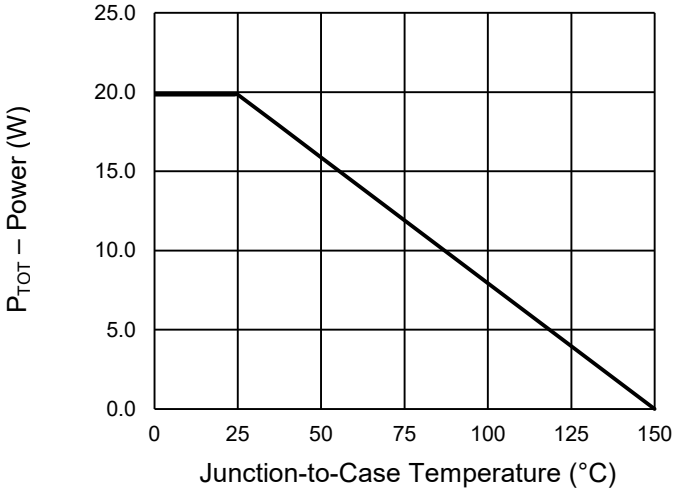
Normalized Thermal Transient Impedance, Junction-to-Cas



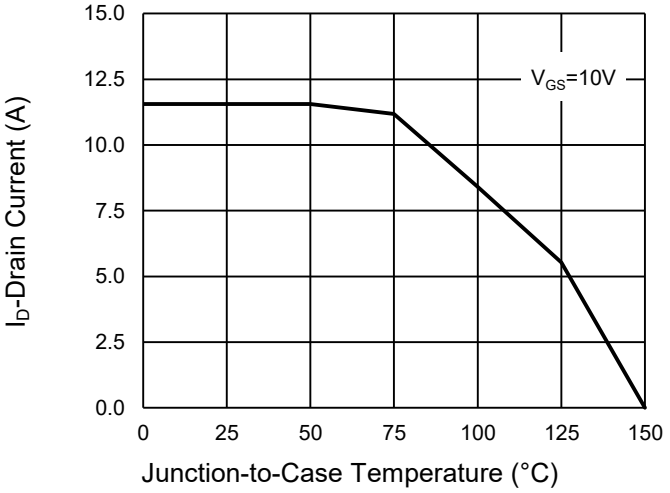
CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

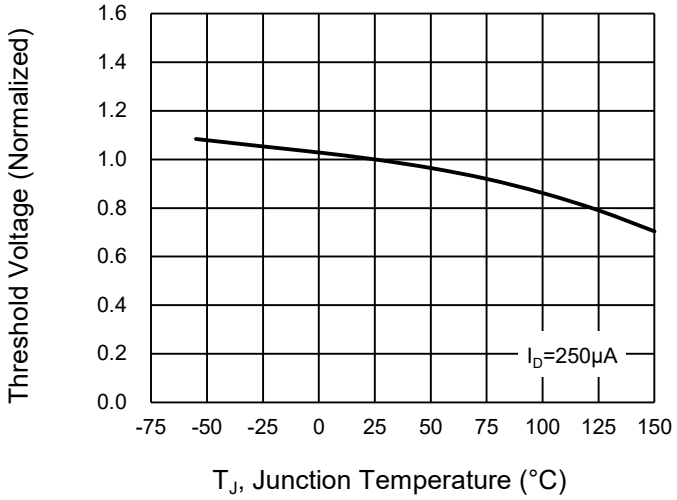
Power Dissipation



Drain Current

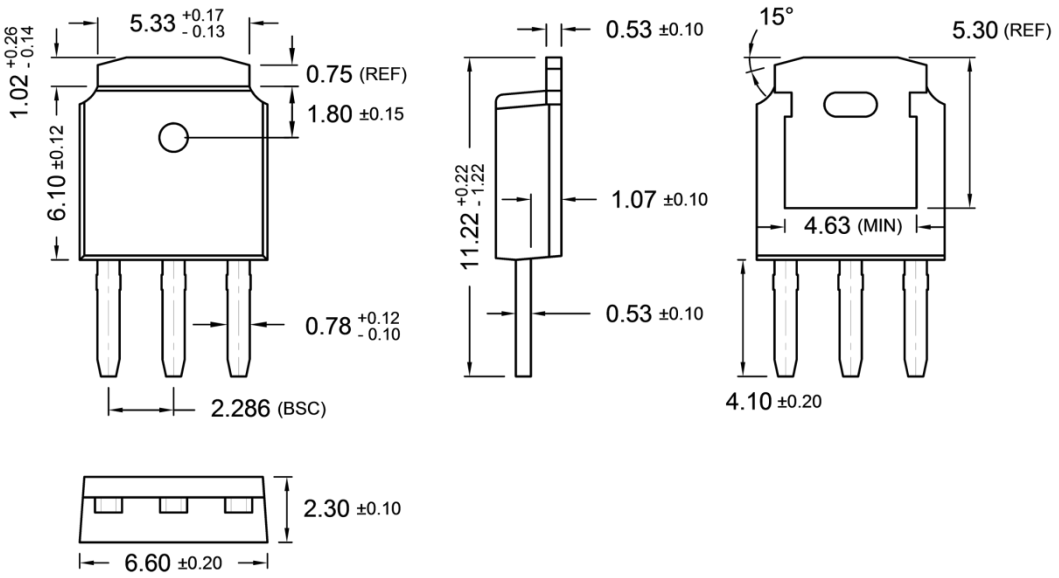


Normalized gate threshold voltage vs Temperature

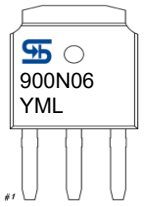


PACKAGE OUTLINE DIMENSIONS (Unit: Millimeters)

TO-251S (IPAK SL)



MARKING DIAGRAM



Y = Year Code

M = Month Code

O	=Jan	**P**	=Feb	**Q**	=Mar	**R**	=Apr
S	=May	**T**	=Jun	**U**	=Jul	**V**	=Aug
W	=Sep	**X**	=Oct	**Y**	=Nov	**Z**	=Dec

L = Lot Code (1~9, A~Z)

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